Annex VI. Debt Sustainability Analysis[1,2]

Lesotho's risk of external debt distress has been revised from "moderate" in the 2015 Article IV to "low" in 2017,[3] mainly due to the recent rebasing of GDP[4], but also to solid GDP growth and revisions to the assessment for new borrowing requirements-the latter tied to non-inclusion of a hydropower project in the current analysis. Under the baseline scenario, all indicators remain below their corresponding thresholds throughout the projection period, with the present value of external public and publicly guaranteed (PPG) debt-to-GDP ratio reaching a peak of around 24 percent by 2020, well below its threshold of 40 percent of GDP. Under stress tests, a large negative shock to exports constitutes the main risk for a possible deterioration of external debt indicators, but even under the most extreme simulated shocks, debt ratios remain below their corresponding thresholds. These findings illustrate that Lesotho has some scope to address challenges from the severe cyclical downswing of Southern African Customs Unions (SACU) revenues by new external borrowing. However, the sensitivity to export shocks illustrates the need for a sizable grant component in new external debt as part of a prudent debt strategy to preserve the low risk of debt distress. Domestic debt is low at around 3 percent of GDP in 2016, with additional domestic contingent liabilities from the underfunded pension system adding about 3.5 percentage points of GDP in present value terms.

A. Recent Developments and Underlying Assumptions

1. Lesotho's total public debt declined from 41.3 percent of GDP in 2015 to 35.4 percent in 2016. This drop is the result of a decline in the valuation of foreign-exchange denominated external debt due to a 10 percent Rand appreciation, limited new borrowing, and solid GDP growth. External debt represented 88.7 percent of total debt, with a substantial share contracted with multilateral creditors. The stock of domestic debt could increase if large pension fund liabilities, estimated at 3.5 percent of GDP in present value terms, remain unfunded, but the government has initiated work with World Bank assistance to address this issue. Technical assistance has been provided in 2017 to assist the government in developing a medium-term debt strategy.

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[1] Prepared by the staffs of the International Monetary Fund (IMF) and the International Development Association (IDA). Approved by Anne Marie Gulde (AFR), Gavin Gray (SPR) and Paloma Anos Casero (IDA).

[2] The DSA updated the previous Join DSA dated January 12, 2016 (IMF Country Report No. 16/33). The DSA follows the IMF and World Bank Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries (November 5, 2013). The DSA uses the unified discount rate of 5 percent set out in Decision No. 15462 (October 11, 2013). For the purpose of defining debt burden thresholds under the Debt Sustainability Framework (DSF), Lesotho is classified as a medium policy performer. Lesotho's average rating on the World Bank's Country Policy and Institutional Assessment (CPIA) for the period 2014-16 is 3.37. The corresponding indicative thresholds are: 40 percent for the NPV of PPG external debt-to-GDP ratio; 150 percent for the net present value (NPV) of debt-to-exports ratio; 250 percent for NPV of debt-to-revenue ratio; 20 percent for the debt service-to-exports ratio; and 20 percent for the debt service-to-revenue ratio.

[3] The years refer to fiscal years running from April 1st.

[4] Rebased national accounts were published in October 2016. The base year is 2012. Nominal GDP at local current prices was revised upwards by about 10 percent on average, from previously reported national accounts.

2. The previous DSA (2015) assessed Lesotho's risk of external debt distress as moderate. The moderate rating was driven by a decision to include financing for a potential hydropower project that could be added to the second phase of the Lesotho Highlands Water Project (LHWP-II)—a mega infrastructure project to export drinking water to South Africa. While LHWP II is grant funded, the hydropower project would probably be financed mostly on commercial terms. Assumptions regarding possible external borrowing associated with this project pushed the PV of PPG external debt-to-GDP, in the 2015 DSA, to a level close to the 40 percent threshold by 2025. Consequently, the stress test yielded a protracted breach, maintaining Lesotho's debt rating at moderate. While there were indications at the 2015 Article IV that this project was clearly specified and the government indicated strong interest in pursuing it, the project turned out more complicated than expected. Therefore, the government initiated new feasibility studies to explore different technical options with very different cost and economic implications. Only once these feasibility studies are completed, the government will be able to decide to pursue the project or not. With discussions still at an early stage, this project is not included in the macroeconomic baseline projections for the 2017 DSA.

Lesotho: Public Outstanding Debt
(As of end-March, 2017)

	US$ million	Percent of total
Domestic debt	100	11.3
External debt	788	88.7
Mulitlateral	662	74.6
IDA	282	31.7
ADF	122	13.8
EIB	116	13.1
IMF	65	7.3
BADEA	32	3.6
IFAD	30	3.4
OFID	15	1.7
Bilateral	120	13.5
China Exim Bank	50	5.6
Kuwait Fund	27	3.1
Saudi Fund	23	2.6
Abu Dhabi Fund	15	1.6
India Exim Bank	6	0.6
Commercial	6	0.6
Total	888	100.0

Source: Country Authorities.

3. Lesotho's public debt sustainability analysis assumes that the fiscal stance will remain loose during the cyclical downturn in SACU revenues. SACU revenues have been well below historical levels and this cyclical downturn will probably last for two years due to the projected slow cyclical recovery in South Africa. After that, SACU revenues are projected to recover in line with projections for South Africa, leading to broadly balanced budgets. The DSA results are based on a macroeconomic baseline scenario, discussed with the authorities during the 2017 Article IV consultation mission, that has embedded no significant fiscal adjustment in response to the SACU revenue shortfall. The DSA is based on the following assumptions:

- **Macroeconomic assumptions.** Real GDP growth is projected to reach 3.1 percent in 2017/18, but ease to 1.7 percent in 2018/19 due to the difficult macroeconomic environment, before benefitting over the medium term from mining and the construction of

the construction of the LHWP II. By 2025, when LHWP II is finalized, the current account deficit is expected to narrow gradually, helped by the additional water exports and weaker capital goods imports. Over the long term, real GDP growth is conservatively projected to average about 3½ percent a year, broadly in line with the previous DSA and below the 4.1 percent growth average over the last 10 years. Annual CPI inflation is expected to remain broadly unchanged at 5.5 percent until 2019/20 and to remain around 5 percent in the medium term. Exports are expected to be mainly driven by diamonds, textiles, and water, while imports incorporate the effects of large investment projects.

Lesotho. Macroeconomic Assumptions (Averages)

	2015 DSA 2015-20	2017 DSA 2017-22	2015 DSA 2021-35	2017 DSA 2023-37
Nominal GDP (US$) million	2317.0	3158.0	4252.0	5661.8
Real GDP Growth (Percent)	3.8	3.1	3.5	3.4
Inflation (Percent average)	5.0	5.5	5.0	5.1
Domestic Primary Deficit (Percent of GDP)	1.2	1.6	-0.6	-0.6
USD Export Growth (Percent)	5.0	7.9	5.8	7.3
USD Import Growth (Percent)	1.3	8.6	5.2	4.6
Non-interest Current Account Balance (Percent of External Debt (Percent of GDP)	-8.5	-8.3	-5.7	-2.1
Public Sector Debt (Percent of GDP)	44.0	34.5	47.0	30.3
	48.4	42.6	52.1	40.8

Sources: IMF Country Report No. 16/33 and IMF and World Bank staffs calculations.

- **Fiscal assumptions.** Fiscal year 2016/17 ended with a deficit of 6.3 percent of GDP. Despite a moderate rise in SACU revenues, the FY 2017/18 deficit will likely reach about 6.5 percent of GDP due to domestic revenue shortfalls. A cyclical upswing in SACU revenues is expected to take place by FY 2020/21 in line with a recovery for South Africa, leading to broadly balanced budgets.

- **Financing assumptions.** The DSA outcomes result from assuming a borrowing plan for the medium term consistent with the need for highly concessional external financing. The large grant element contained in the borrowing of recent years is expected to continue in the medium term. With 75 percent of total outstanding debt owed to multilateral creditors carrying a large concessional component, the 2017 DSA assumes that the grant element in the new borrowing is expected to remain high.

B. External Debt Sustainability Analysis

Baseline

4. All external debt sustainability indicators remain below their corresponding thresholds in the baseline scenario (Table and Figure 1). The present value (PV) of PPG external debt-to-GDP is expected to reach a maximum of 23.6 percent by 2020. The debt path is supported by a large grant element assumed in the new borrowing (42 percent) in line with past borrowing practices. In addition to the ongoing projects and their implied disbursements, the government is planning to borrow on average US$45 to 50 million per year to finance new projects in the medium term. The

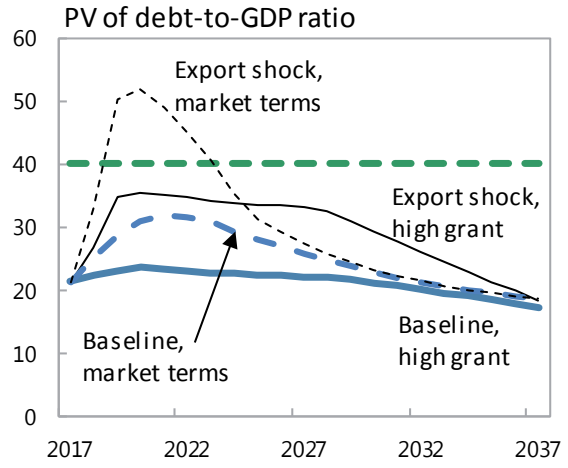
new financing is a mix of foreign development assistance and partly concessional loans (IDA, African Development Bank, EU, Saudi Fund, Abu Dhabi, BADEA and China), aimed at projects to cover the expansion of the airport, roads (Mpiti-Sehlabathebe), water supply (Lowlands Water Project), tax modernization, and other infrastructure projects (Mesitsaneng-Ha TSepo). In addition, Lesotho's eligibility for the Millennium Challenge Corporation, which was approved in December 2017, should open up access to significant grant financing.

Sensitivity Analysis

5. Stress tests show that Lesotho's external debt vulnerabilities would increase significantly, in particular in the event of a major shock to exports, like a terms-of-trade shock (Table 2). In a scenario where export values grow at one standard deviation lower than the historical average (terms-of-trade shock), the shock would likely bring the PV of debt-to-GDP ratio close to the 40 percent threshold in 2020, and remain near threshold for a few years. An export shock would also carry adverse effects to other main external debt indicators, but without breaching the relevant thresholds. The PV of external debt-to-export ratio would peak at 120.2 percent by 2019. In a scenario in which the key variables are set at their average of the past 10 years, Lesotho's external debt ratios fall relative to the baseline, reflecting an average fiscal surplus over the past years thanks to large SACU revenues, strong GDP growth and inflation averaging about 5 percent. However, given the structural breaks in Lesotho's economy, especially the cyclical character of the currently very low SACU revenues, the historical scenario could be considered less relevant for the analysis. In addition, the LHWP II will increase exports once finalized by the end of the medium term.

6. The risk of debt distress is sensitive to the assumption regarding the concessionality of new borrowing, which should be defined in the government as part of a new debt strategy. The text figure highlights differences in debt indicator outturns should new borrowing be transacted at similar terms to that of the current portfolio, vis-à-vis that the current portfolio is replaced by new borrowing at market terms. For the latter, the PV of debt-to-GDP would reach a maximum of 32 percent, still below the threshold. However, under a shock to exports the threshold would be breached (See Figure). This underlines the urgency for the government to develop a medium-term debt strategy that defines the appropriate financing mix.

Lesotho. Impact of Lower Grant on External Debt



C. Public Sector Debt Sustainability

7. Domestic debt is relatively small, and the total public sector debt trajectory is largely determined by external debt developments. A continuous, moderate increase in domestic debt will help finance the projected fiscal deficits in the coming years. Domestic debt would remain at 4 to 5 percent of GDP throughout the projected period unless contingent liabilities from the pension fund materialize.

8. The PV of public sector debt stood at 24.6 percent of GDP in 2016 (Table 3). The standard sensitivity tests for public debt distress point to a higher vulnerability to a real depreciation of the exchange rate (Table 4), pushing the ratio upward but below its benchmark for the forecast period. Adequate buffers would help to keep stressed debt indicators well below the critical levels. The inclusion of contingent liabilities from unfunded pension funds (about 3.5 percent in present value terms) would increase the public debt ratios, but they remain below the indicative benchmark. While guarantees for external debt are monitored well by the Ministry of Finance, weak monitoring of state-owned enterprises limits information about possible domestic contingent liabilities which are, therefore, not included in the DSA. However, based on World Bank analysis, the range of possible contingent liabilities would not have a major impact on the DSA.

9. The fixed primary balance simulation leads to a sharp increase in public debt. The upward trend in public debt under this scenario is driven by the large cyclical upswing in the primary fiscal deficit estimated for 2017. In this simulation, the deficit is kept unchanged at 5.7 percent of GDP for the whole forecast period, generating very large financing needs that continue to grow over time. In contrast, a more realistic assumption is reflected in the baseline, including a gradual recovery in South Africa's GDP which is the major determinant of SACU revenues.

D. Conclusion

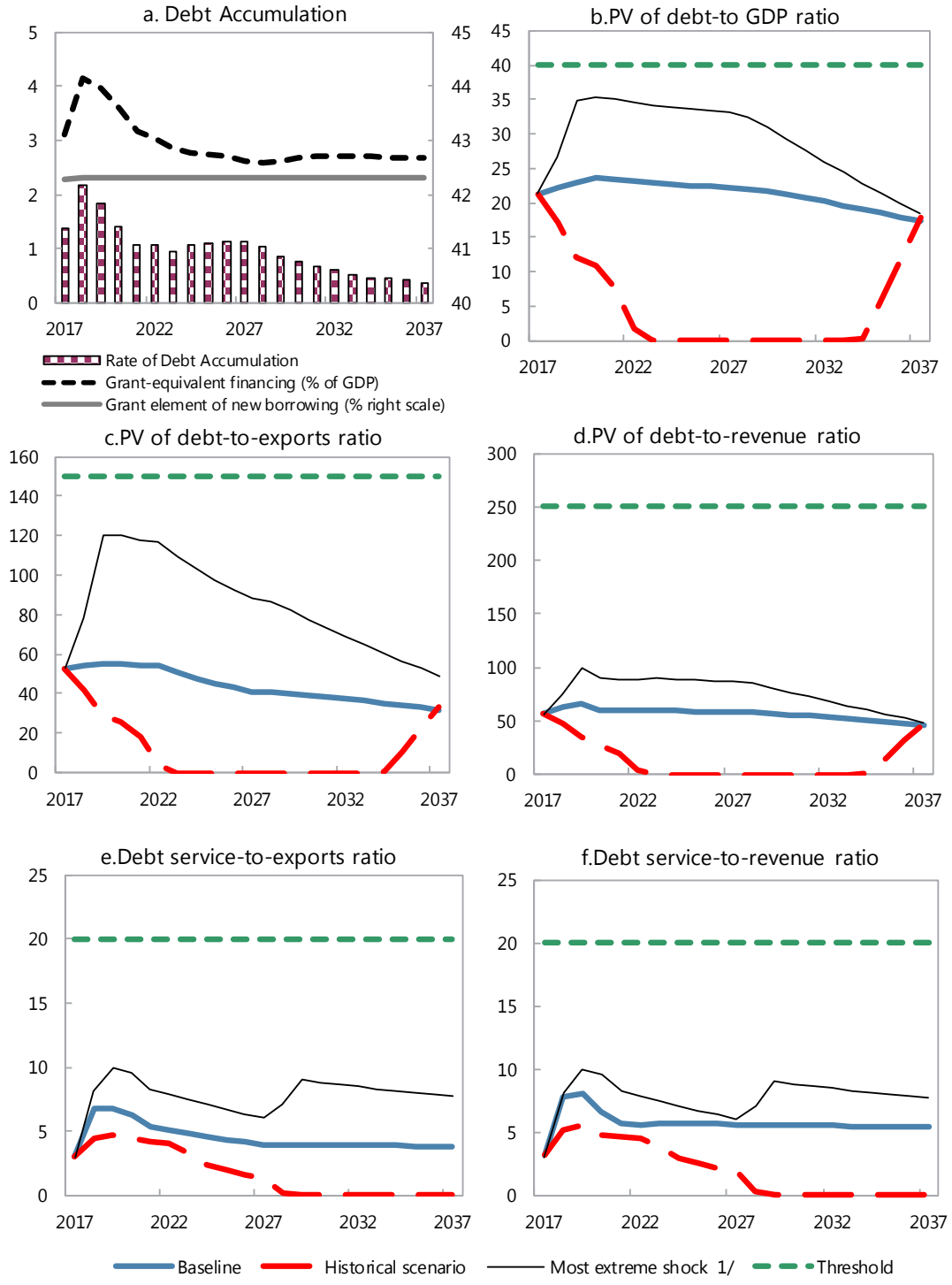
10. Lesotho's risk of debt distress has been revised from moderate to low in 2017. All external debt and debt service indicators for the baseline and the simulated shocks remain below their respective thresholds. A large shock to exports would pose the larger risk to debt sustainability, but even the most extreme shocks do not bring the debt ratios above the threshold.

11. These simulations suggest that there is some —though limited— scope for additional non-concessional borrowing to finance large infrastructure projects with large economic returns. Careful cost-benefit analysis and debt sustainability implications must be considered in future government investment plans, including discussions of the large hydropower project for which feasibility studies are under way. Maintaining the significant grant element in the financing will avoid risks for future fiscal deficit and international reserves. The sensitivity to financing assumptions also underlines the need for the government to prepare a medium-term debt strategy.

Authorities views

12. Authorities concurred with the DSA analysis and the "low" risk rating. They agreed with the need to better monitor domestic contingent liabilities to have a comprehensive view of the debt. They also agreed that prudent debt management must continue in the medium term, in particular by pursuing financing with a significant grant element.

Figure 1. Lesotho: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2014–37[1]



Sources: Country authorities and IMF staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio on or before 2027. In figures b, c, d, e, and f it corresponds to an exports shock.

Table 1. Lesotho: Public and Publicly Guaranteed External Debt Sustainability Framework, Baseline Scenario, 2014—37 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**34.3**	**38.3**	**32.3**			**31.4**	**33.3**	**34.9**	**35.9**	**35.7**	**35.5**		**32.7**	**25.2**	
of which: public and publicly guaranteed (PPG)	34.3	38.3	32.3			31.4	33.3	34.9	35.9	35.7	35.5		32.7	25.2	
Change in external debt	-0.2	3.9	-6.0			-0.9	1.9	1.5	1.1	-0.2	-0.3		-0.6	-0.7	
Identified net debt-creating flows	-1.1	4.3	2.9			3.6	8.8	8.8	1.6	3.7	8.0		-2.6	-4.7	
Non-interest current account deficit	**4.5**	**3.9**	**6.8**	**0.9**	**11.4**	**6.0**	**10.8**	**11.1**	**3.9**	**6.7**	**11.1**		**0.3**	**-1.9**	2.1
Deficit in balance of goods and services	47.9	43.1	39.8			43.0	44.8	42.9	39.9	42.2	46.0		22.9	22.9	
Exports	34.1	41.8	39.9			40.3	41.0	41.7	42.4	42.7	42.8		54.1	54.1	
Imports	82.0	84.8	79.7			83.3	85.9	84.6	82.3	84.9	88.8		77.0	77.0	
Net current transfers (negative = inflow)	-30.8	-26.1	-19.3	-30.7	6.8	-23.4	-20.5	-18.2	-22.5	-22.1	-21.7		-22.3	-24.6	-23.0
of which: official	-25.7	-21.2	-14.4			-18.5	-15.6	-13.3	-17.6	-17.2	-16.8		-15.8	-12.9	
Other current account flows (negative = net inflow)	-12.5	-13.1	-13.6			-13.5	-13.5	-13.5	-13.5	-13.4	-13.2		-0.4	-0.2	
Net FDI (negative = inflow)	**-4.2**	**-4.5**	**-3.1**	**-1.8**	**2.5**	**-2.0**	**-2.0**	**-2.0**	**-2.0**	**-2.0**	**-2.0**		**-2.2**	**-2.2**	-2.1
Endogenous debt dynamics 2/	**-1.4**	**5.0**	**-0.8**			**-0.4**	**0.0**	**-0.3**	**-0.2**	**-1.0**	**-1.1**		**-0.7**	**-0.7**	
Contribution from nominal interest rate	0.5	0.6	0.6			0.5	0.5	0.5	0.5	0.5	0.5		0.4	0.2	
Contribution from real GDP growth	-1.0	-1.0	-1.1			-0.9	-0.5	-0.8	-0.7	-1.5	-1.6		-1.1	-0.9	
Contribution from price and exchange rate changes	-0.9	5.3	-0.3			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**0.9**	**-0.4**	**-8.9**			**-4.5**	**-6.9**	**-7.3**	**-0.6**	**-3.9**	**-8.3**		**2.0**	**4.0**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	21.8			21.3	22.3	23.0	23.6	23.3	23.1		22.2	17.3	
In percent of exports	...	...	54.7			52.9	54.3	55.2	55.5	54.7	54.0		41.1	32.0	
PV of PPG external debt	**...**	**...**	**21.8**			**21.3**	**22.3**	**23.0**	**23.6**	**23.3**	**23.1**		**22.2**	**17.3**	
In percent of exports	**...**	**...**	**54.7**			**52.9**	**54.3**	**55.2**	**55.5**	**54.7**	**54.0**		**41.1**	**32.0**	
In percent of government revenues	**...**	**...**	**56.7**			**56.5**	**62.3**	**65.7**	**59.4**	**59.4**	**59.3**		**58.3**	**45.4**	
Debt service-to-exports ratio (in percent)	**4.1**	**4.4**	**4.9**			**3.0**	**6.8**	**6.8**	**6.2**	**5.3**	**5.1**		**4.0**	**3.8**	
PPG debt service-to-exports ratio (in percent)	**4.1**	**4.4**	**4.9**			**3.0**	**6.8**	**6.8**	**6.2**	**5.3**	**5.1**		**4.0**	**3.8**	
PPG debt service-to-revenue ratio (in percent)	**2.9**	**4.2**	**5.1**			**3.2**	**7.8**	**8.1**	**6.7**	**5.8**	**5.6**		**5.6**	**5.4**	
Total gross financing need (Millions of U.S. dollars)	268.8	240.6	291.3			256.4	459.6	493.6	274.2	371.2	545.7		213.6	185.8	
Non-interest current account deficit that stabilizes debt ratio	4.7	-0.1	12.8			6.9	8.9	9.6	2.8	6.9	11.4		0.9	-1.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.0	2.5	3.1	4.1	1.7	3.1	1.7	2.6	2.1	4.4	4.7	3.1	3.5	3.5	3.4
GDP deflator in US dollar terms (change in percent)	2.6	-13.3	0.8	-0.7	10.6	10.0	4.4	2.4	1.8	1.1	0.8	3.4	2.0	2.0	2.1
Effective interest rate (percent) 5/	1.4	1.7	1.7	1.4	0.2	1.7	1.7	1.5	1.5	1.4	1.4	1.5	1.1	0.8	1.0
Growth of exports of G&S (US dollar terms, in percent)	6.1	8.8	-0.9	1.8	12.2	14.6	8.1	6.7	5.7	6.2	5.8	7.9	9.8	5.6	7.3
Growth of imports of G&S (US dollar terms, in percent)	1.3	-8.0	-2.5	0.6	11.9	18.5	9.4	3.4	1.2	8.8	10.5	8.6	3.5	5.6	4.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	42.3	42.3	42.3	42.3	42.3	42.3	42.3	42.3	42.3	42.3
Government revenues (excluding grants, in percent of GDP)	48.1	44.2	38.4			37.7	35.8	35.0	39.6	39.3	38.9		38.1	38.1	38.1
Aid flows (in Millions of US dollars) 7/	123.8	78.8	73.3			84.4	35.0	45.6	41.6	46.5	57.7		53.3	43.3	
of which: Grants	44.8	70.1	61.6			58.3	60.3	61.6	62.6	63.3	63.9		83.7	144.0	
of which: Concessional loans	79.0	8.7	11.7			26.1	-25.4	-16.0	-21.0	-16.7	-6.2		-30.4	-100.7	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.1	4.1	4.0	3.6	3.2	3.1		2.6	2.7	2.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			69.6	59.3	59.7	61.6	64.0	63.8		69.8	68.8	68.5
Memorandum items:															
Nominal GDP (Millions of US dollars)	2646.6	2350.9	2442.0			2768.1	2937.6	3083.9	3205.3	3381.8	3571.0		4682.7	8051.9	
Nominal dollar GDP growth	5.6	-11.2	3.9			13.4	6.1	5.0	3.9	5.5	5.6	6.6	5.6	5.6	5.6
PV of PPG external debt (in Millions of US dollars)			563.7			597.3	657.9	712.0	755.7	790.0	825.6		1041.6	1394.0	
(PVt-PVt-1)/GDPt-1 (in percent)						1.4	2.2	1.8	1.4	1.1	1.1	1.5	1.1	0.4	0.8
Gross workers' remittances (Millions of US dollars)	183.3	155.4	153.2			166.8	242.1	270.9	251.3	296.4	299.5		412.7	784.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	20.5			20.1	20.6	21.2	21.8	21.5	21.3		20.4	15.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	47.2			46.0	45.2	45.6	46.8	45.4	45.1		35.3	27.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	4.2			2.6	5.6	5.6	5.3	4.4	4.3		3.4	3.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37[1/]

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	21	22	23	24	23	23	**22**	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	21	17	12	11	8	2	**0**	18
A2. New public sector loans on less favorable terms in 2017-2037 2/	21	20	19	17	14	11	**-3**	-32
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	21	22	23	23	23	23	**22**	17
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	21	27	35	35	35	35	**33**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	21	26	31	32	32	31	**30**	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	21	22	21	21	21	21	**20**	17
B5. Combination of B1-B4 using one-half standard deviation shocks	21	26	32	33	33	32	**31**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	21	31	32	33	33	32	**31**	24
PV of debt-to-exports ratio								
Baseline	53	54	55	56	55	54	**41**	32
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	53	42	29	26	18	4	**0**	33
A2. New public sector loans on less favorable terms in 2017-2037 2/	53	49	45	40	33	27	**-5**	-58
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	53	55	55	56	55	54	**41**	32
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	53	78	120	120	118	116	**88**	49
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	53	55	55	56	55	54	**41**	32
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	53	54	50	50	50	49	**37**	32
B5. Combination of B1-B4 using one-half standard deviation shocks	53	65	82	83	81	80	**61**	41
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	53	55	55	56	55	54	**41**	32
PV of debt-to-revenue ratio								
Baseline	57	62	66	59	59	59	**58**	45
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	57	48	35	28	20	5	**0**	47
A2. New public sector loans on less favorable terms in 2017-2037 2/	57	57	54	42	36	29	**-8**	-83
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	57	62	66	59	59	59	**58**	45
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	57	74	100	89	89	89	**87**	48
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	57	74	90	81	81	81	**79**	62
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	57	61	60	54	54	54	**53**	45
B5. Combination of B1-B4 using one-half standard deviation shocks	57	72	92	83	83	83	**81**	55
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	57	88	93	84	83	83	**82**	64

Table 2. Lesotho: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37 (continued)

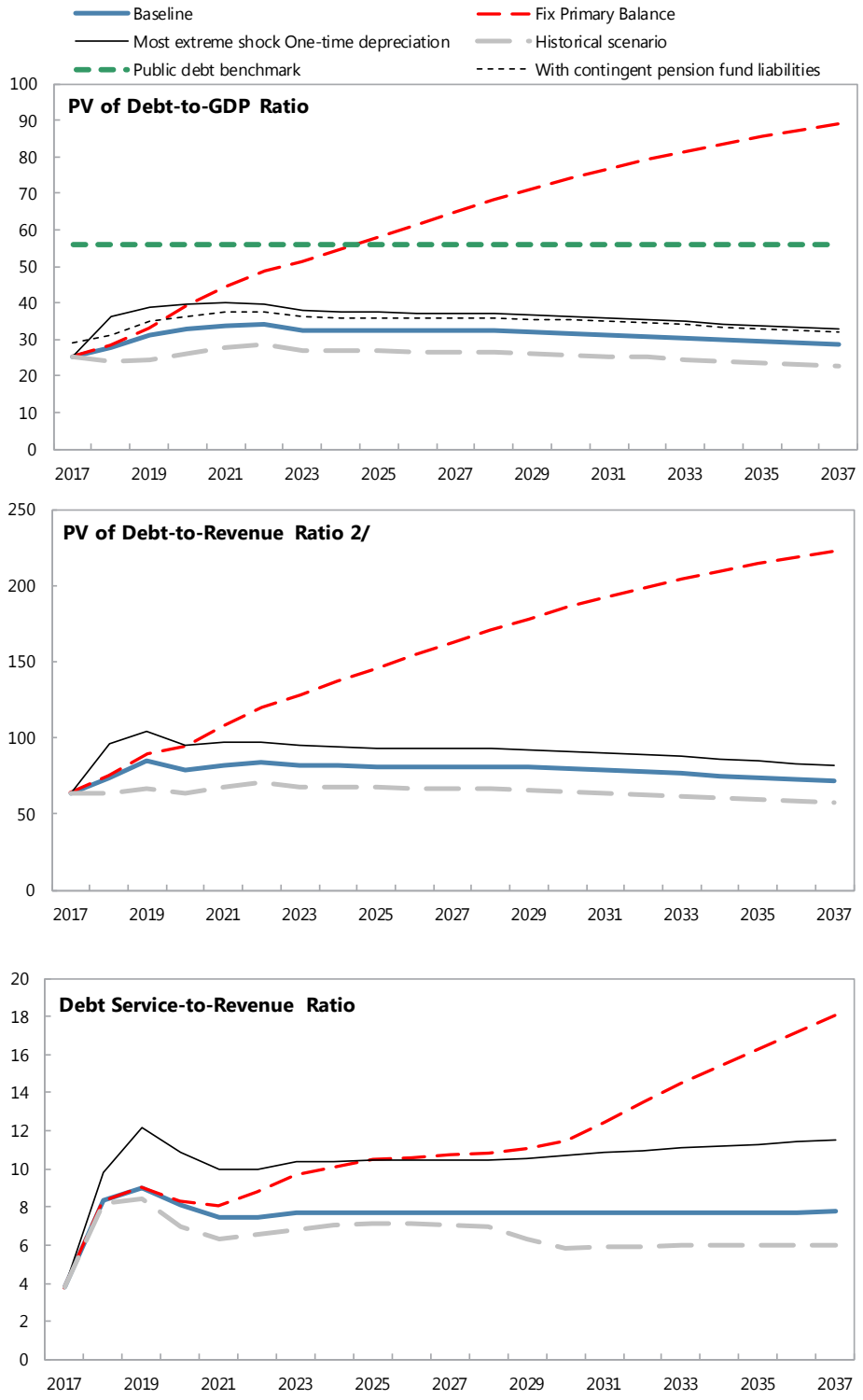
(In percent)

Debt service-to-exports ratio								
Baseline	3	7	7	6	5	5	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	3	5	5	4	4	4	**1**	0
A2. New public sector loans on less favorable terms in 2017-2037 2/	3	4	5	4	4	4	**1**	-3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	3	7	7	6	5	5	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	3	8	10	10	8	8	**6**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	3	7	7	6	5	5	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	3	7	7	6	5	5	**4**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	3	8	9	8	7	7	**5**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	3	7	7	6	5	5	**4**	4
Debt service-to-revenue ratio								
Baseline	3	8	8	7	6	6	**6**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	3	5	6	5	5	4	**2**	0
A2. New public sector loans on less favorable terms in 2017-2037 2/	3	5	5	5	4	4	**2**	-4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	3	7	7	6	5	5	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	3	8	10	10	8	8	**6**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	3	7	7	6	5	5	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	3	7	7	6	5	5	**4**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	3	8	9	8	7	7	**5**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	3	7	7	6	5	5	**4**	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	40	40	40	40	40	40	**40**	40

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 2. Lesotho: Indicator of Public Debt Under Alternative Scenarios, 2017–37[1/]



Sources: Country auhtorities and IMF staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.

Table 3. Lesotho: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014–37

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	36.9	41.3	35.4			35.4	38.8	43.2	45.3	46.2	46.4		42.9	36.3	
of which: foreign-currency denominated	34.3	38.3	32.3			31.4	33.3	34.9	35.9	35.7	35.5		32.7	25.2	
Change in public sector debt	-0.5	4.4	-5.8			0.0	3.4	4.4	2.2	0.9	0.2		-0.5	-0.6	
Identified debt-creating flows	-0.7	4.0	0.4			3.7	3.5	2.9	-2.3	-2.8	-3.0		-1.9	-1.6	
Primary deficit	-1.0	0.2	5.4	-0.6	6.1	5.7	4.5	3.9	-1.7	-1.5	-1.5	1.6	-0.7	-0.6	-0.6
Revenue and grants	49.8	47.2	40.9			39.8	37.8	37.0	41.6	41.2	40.7		39.9	39.9	
of which: grants	1.7	3.0	2.5			2.1	2.1	2.0	2.0	1.9	1.8		1.8	1.8	
Primary (noninterest) expenditure	48.8	47.4	46.4			45.5	42.3	40.9	39.9	39.7	39.2		39.3	39.3	
Automatic debt dynamics	0.3	3.8	-5.0			-2.0	-1.0	-0.9	-0.6	-1.4	-1.5		-1.3	-1.0	
Contribution from interest rate/growth differential	-1.3	-0.7	-0.8			-1.0	-0.6	-0.9	-0.8	-1.7	-1.9		-1.3	-1.0	
of which: contribution from average real interest rate	-0.2	0.2	0.4			0.0	0.0	0.1	0.1	0.2	0.2		0.2	0.2	
of which: contribution from real GDP growth	-1.1	-0.9	-1.2			-1.1	-0.6	-1.0	-0.9	-1.9	-2.1		-1.5	-1.3	
Contribution from real exchange rate depreciation	1.5	4.5	-4.2			-0.9	-0.4	0.0	0.2	0.3	0.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.3	0.4	-6.3			-3.7	-0.1	1.4	4.4	3.7	3.2		1.4	1.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	24.9			25.3	27.8	31.3	33.0	33.8	34.0		32.4	28.5	
of which: foreign-currency denominated	...	...	21.8			21.3	22.3	23.0	23.6	23.3	23.1		22.2	17.3	
of which: external	...	...	21.8			21.3	22.3	23.0	23.6	23.3	23.1		22.2	17.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	0.7	2.3	7.7			7.2	7.6	7.2	1.7	1.6	1.5		2.4	2.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	60.9			63.7	73.4	84.6	79.2	82.2	83.6		81.2	71.4	
PV of public sector debt-to-revenue ratio (in percent)	...	...	64.9			67.3	77.6	89.5	83.1	86.1	87.4		85.1	74.7	
of which: external 3/	...	...	56.7			56.5	62.3	65.7	59.4	59.4	59.3		58.3	45.4	
Debt service-to-revenue and grants ratio (in percent) 4/	3.3	4.4	5.5			3.8	8.3	9.0	8.1	7.5	7.5		7.7	7.7	
Debt service-to-revenue ratio (in percent) 4/	3.4	4.7	5.8			4.0	8.8	9.5	8.5	7.8	7.8		8.1	8.1	
Primary deficit that stabilizes the debt-to-GDP ratio	-0.5	-4.2	11.3			5.7	1.1	-0.5	-3.8	-2.3	-1.7		-0.1	0.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.0	2.5	3.1	4.1	1.7	3.1	1.7	2.6	2.1	4.4	4.7	3.1	3.5	3.5	3.4
Average nominal interest rate on forex debt (in percent)	1.4	1.7	1.7	1.4	0.2	1.7	1.7	1.5	1.5	1.4	1.4	1.5	1.1	0.8	1.0
Average real interest rate on domestic debt (in percent)	-2.6	1.2	7.6	3.1	3.9	3.4	3.9	4.3	3.9	4.3	3.9	3.9	4.8	4.8	4.6
Real exchange rate depreciation (in percent, + indicates depreciation)	4.6	13.2	-11.2	1.5	13.4	-3.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	12.0	8.0	2.8	6.6	3.6	6.5	5.9	5.5	5.3	5.0	4.9	5.5	5.0	5.0	5.1
Growth of real primary spending (deflated by GDP deflator, in percent)	-5.6	-0.5	0.8	-0.5	1.9	1.1	-5.4	-0.9	-0.3	3.8	3.5	0.3	3.5	3.5	3.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	42.3	42.3	42.3	42.3	42.3	42.3	42.3	42.3	42.3	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Lesotho: Sensitivity Analysis for Key Indicators of Public Debt, 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	25	28	31	33	34	34	32	28
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	24	24	26	28	29	27	23
A2. Primary balance is unchanged from 2017	25	29	33	39	45	49	65	89
A3. Permanently lower GDP growth 1/	25	28	32	34	35	36	37	44
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	25	27	31	32	33	33	30	24
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	25	28	33	35	35	36	34	29
B3. Combination of B1-B2 using one half standard deviation shocks	25	26	29	30	30	30	26	18
B4. One-time 30 percent real depreciation in 2018	25	36	39	40	40	40	37	33
B5. 10 percent of GDP increase in other debt-creating flows in 2018	25	34	37	39	40	40	38	32
PV of Debt-to-Revenue Ratio 2/								
Baseline	64	73	85	79	82	84	81	71
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	64	64	66	63	67	71	67	57
A2. Primary balance is unchanged from 2017	64	75	90	95	108	120	162	223
A3. Permanently lower GDP growth 1/	64	74	86	81	85	87	93	109
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	64	72	83	77	79	80	75	61
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	64	75	89	83	86	87	85	74
B3. Combination of B1-B2 using one half standard deviation shocks	64	69	77	71	73	73	64	46
B4. One-time 30 percent real depreciation in 2018	64	96	104	95	97	97	93	82
B5. 10 percent of GDP increase in other debt-creating flows in 2018	64	90	101	94	97	98	95	80
Debt Service-to-Revenue Ratio 2/								
Baseline	4	8	9	8	7	7	8	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4	8	8	7	6	7	7	6
A2. Primary balance is unchanged from 2017	4	8	9	8	8	9	11	18
A3. Permanently lower GDP growth 1/	4	8	9	8	8	8	8	10
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	4	8	9	8	7	7	7	7
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	4	8	9	8	8	8	8	8
B3. Combination of B1-B2 using one half standard deviation shocks	4	8	9	8	7	7	7	6
B4. One-time 30 percent real depreciation in 2018	4	10	12	11	10	10	10	12
B5. 10 percent of GDP increase in other debt-creating flows in 2018	4	8	10	9	8	8	8	9

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.